

ANUBHAV SEN

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EDUCATION

Bachelor of Science in Computer Engineering

The Pennsylvania State University, University Park, PA
College of Engineering | Schreyer Honors College

Graduation: May 2026
Cumulative GPA: **3.94/4.0**

TECHNICAL SKILLS

- Proficient: C, Java, Python, Dash, Git, Android SDK, Android Studio, Google Firebase, XML, Verilog
- Intermediate: SQL, JavaScript, Node.js, MongoDB, React Native, Shiny

WORK EXPERIENCE

Global Data Operations Intern @ Merck & Co.

June 2024 – August 2024

- Developed a streamlined, user-friendly dashboard to centralize data automation tools, significantly improving accessibility and usability while **reducing time spent on manual tasks by 30%** for **500+ users globally**.
- Created a one-click interface for running these tools, garnering strong support from non-technical users and eliminating the need for any prior experience using the command-line shell or knowing system-level commands.
- Incorporated Dash framework and collaborated with IT to secure access for deploying the site on AWS servers.

Computer Science Grader & Learning Assistant (LA)

January 2023 – August 2025

- Coached **300+ students** on data structures + algorithms (e.g., complexity analysis, graph theory, DP, LP, etc.).
- Facilitated 2 recitations weekly and provided 1:1 tutoring during office hours to help students debug their code.

SOFTWARE DEVELOPMENT PROJECTS

React Native Development

May 2023 – Present

- [Nittany Retails](#): e-commerce (JavaScript-based) mobile application **August 2023**
 - Hosted Node.js backend server and connected it to MongoDB for flexible, schema-less data storage.
 - Incorporated Stripe API to handle payment processing in order to generate internal revenue within app.

Android Application Development

August 2019 – Present

- Published **3 Android (Java-based) mobile applications** to Google Play Store that have a growing user base while currently exploring new ways to boost engagement and improve existing features based on user feedback.
- [Smart Planner](#): digital student planner application **July 2022**
 - Implemented a student planning tool with automated reminders and calendar syncing (via the Google Calendar API), giving students a way to easily manage course assignments and deadlines across devices.
- [TalkZone](#): video-calling/casting media application **October 2021**
 - Engineered real-time video calling and cross-device media casting, combining an open-source video conferencing SDK with the native Google Cast API to extend playback from mobile devices to TV screens.
- [TagOut!](#): social media application **July 2021**
 - Developed key features such as phone-based authentication, live messaging, and personalized user feeds, powered by Google Firebase's real-time database and cloud functions for reliable performance at scale.

Video Game Development

August 2018 – August 2019

- Built clones of classical arcade games like *Angry Birds*, *Flappy Bird*, *Snake*, and *Tetris* using Pygame.

RESEARCH EXPERIENCE

Improving circRNA Detection: A Study of Paired-End Read Patterns

February 2025 – Present

- Implementing a circular RNA assembly framework that jointly leverages chimeric and outward reads to more accurately identify back-splicing junctions, overcoming blind spots in tools that rely on either read type alone.

Theory and A Heuristic for the Minimum Path Flow Decomposition Problem

July 2023 – January 2025

- Generalized a minimum path flow decomposition model from directed acyclic graphs to the cyclic case, where the objective was to represent a given edge flow as a weighted sum of the fewest possible source-to-sink paths.

AI for Politics

November 2022 – May 2024

- Predicted future US election outcomes by scraping 10K tweets/day (X API) and training a logistic-regression text classifier using TF-IDF features to label political leaning, then aggregated state-level sentiment into a heat map.
- Assessed performance by testing predictions against prior election results, giving **88% accuracy** as a whole.

LEADERSHIP/ACTIVITIES

AlgoPSU Director

January 2023 – May 2026

- Mentored **25 students** on abstract data structures and walk through LeetCode questions to reinforce concepts.

Java: A Comprehensive Guide (Parts I & II)

October 2021 – August 2022

- Authored and published a two-part written guide giving a primer on the fundamentals and advanced theories of Java; only available on [Amazon](#) as a Kindle eBook, hardcover, and paperback.